ITDS 124 Mathematics of Computing

Chapter 2.3
Part 1: Rules of Inference

กฎการอนุมาน

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Note on this lecture

- This lecture has been developed from several textbooks including
  - Discrete Mathematical Structures by Bernard Kolman, Robert Busby, et al.
  - Discrete Mathematics and Its Applications by Kenneth Rosen
  - Discrete Mathematics: An Open Introduction by Oscar Levin

Proofs (การพิสูจน์ทางคณิตศาสตร์)

- Proofs in mathematics are valid arguments that establish the truth of mathematical statements.
- By an **argument** (การอ้างเหตุผล), we mean a sequence of statements that end with a conclusion.
- By **valid** (ความสมเหตุสมผล), we mean that the conclusion, or final statement of the argument, must follow from the truth of the preceding statements, or **premises** (ข้ออ้าง,สมมติฐาน), of the argument.
- That is, an **argument** is **valid** if and only if it is impossible for all the **premises** to be true and the conclusion to be false.

Before we study mathematical proofs, we will look at…

1. **Arguments** that involve only compound propositions and what it means for an argument involving compound propositions to be valid.
2. A collection of **rules of inference** in propositional logic. These rules of inference are among the most important ingredients in producing valid arguments.
3. Some common forms of incorrect reasoning, called **fallacies**, which lead to invalid arguments.
4. Rules of inference for quantified statements and description of how these rules of inference can be used to produce valid arguments.
5. How rules of inference for propositions and for quantified statements can be combined and used together in complicated arguments.

Valid Arguments

- Consider the following argument involving propositions :

  "If you have a current password, then you can log onto the network."

  "You have a current password."

  Therefore,

  "You can log onto the network."

Premises

Conclusion

- We would like to determine whether this is a **valid** argument.
  - That is, we would like to determine whether the conclusion "You can log onto the network" must be true when the premises "If you have a current password, then you can log onto the network" and "You have a current password" are both true.

Valid Arguments in Propositional Logic (1)

- Using propositional variables, we can rewrite the previous argument in another form.
- Let p to represent "You have a current password"

  q to represent "You can log onto the network."

Then, the argument has the form

$$p \rightarrow q$$
$$p$$
$$\therefore q$$

Valid?

Premises

Conclusion

where $\therefore$ is the symbol that denotes "therefore."

Valid Arguments in Propositional Logic (2)

- We know from our last lecture slide (Theorem 4), the statement

  $((p \rightarrow q) \wedge p) \rightarrow q$ is a tautology.

- In particular, when both $p \rightarrow q$ and p are true, we know that q must also be true.

- We say this **form of argument** is **valid** because whenever all its premises are true, the conclusion must also be true.

Valid Arguments with false conclusion

- Valid argument may have the conclusion to be false if one of the premises is false.

  "If you have a current password, then you can log onto the network."

  "You have a current password."

  "You can log onto the network."

- If the first premise is false, then "If you have a current password" must be true and "you can log onto the network" must be false.
- Then the conclusion "you can log onto the network" is false.

Argument vs Argument form

- **An argument** in propositional logic is a sequence of propositions.
  - All but the final proposition in the argument are called **premises** and the final proposition is called the **conclusion**.
  - An argument is **valid** if the truth of all its premises implies that the conclusion is true.
- **An argument form** in propositional logic is a sequence of compound propositions involving propositional variables.
  - An argument form is **valid** no matter which particular propositions are substituted for the propositional variables in its premises, the conclusion is true if the premises are all true.

Two different arguments, same argument form

"If you have a current password, then you can log onto the network."

"You have a current password."

"You can log onto the network."

"If you have access to the network, then you can change your grade."

"You have access to the network."

"You can change your grade."

$p \rightarrow q$

p

$\therefore q$

Checking the validity of an argument?

- An argument is **valid** if the truth of all its premises implies that the conclusion is true.
- The argument form with premises $p_1, p_2, \ldots, p_n$ and conclusion q is valid, when $(p_1 \wedge p_2 \wedge \cdots \wedge p_n) \rightarrow q$ is a tautology.
- The key to showing that an argument in propositional logic is valid is to show that its argument form is valid.
- So we need techniques to show that argument forms are valid.

Technique #1 (Use you own reasoning)

- Assume you don't remember tautologies and have no easy way to search for those. So you must use your own reasoning to check that if all premises are true then the conclusion must be true.

$$\begin{array}{l} p \\ p \rightarrow q \\ \hline \therefore q \end{array}$$

Examples

- Check the validity of these arguments.

$$\begin{array}{l} \sim q \\ p \rightarrow q \\ \hline \therefore \sim p \end{array}$$

$$\begin{array}{l} p \\ \hline \therefore p \vee q \end{array}$$

$$\begin{array}{l} p \rightarrow q \\ q \\ \hline \therefore p \end{array}$$

$$\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ r \\ \hline \therefore p \vee q \end{array}$$

Technique #2 (Use truth table)

- We can always use a truth table to show that an argument form is valid.
- We do this by showing that
  - whenever the premises are true, the conclusion must also be true. OR
  - $(p_1 \wedge p_2 \wedge \cdots \wedge p_n) \rightarrow q$ is a tautology

$$\begin{array}{l} p \\ p \rightarrow q \\ \hline \therefore q \end{array}$$

p	q	$p \rightarrow q$	$(p \wedge (p \rightarrow q)) \rightarrow q$
T	T	T	T
T	F	F	T
F	T	T	T
F	F	T	T

Examples

- Check the validity of these arguments using the truth table.

$\sim q$

$p \rightarrow q$

$\therefore \sim p$

p	q	$\sim p$	$\sim q$	$p \rightarrow q$	$(\sim q \wedge (p \rightarrow q)) \rightarrow \sim p$

$p \rightarrow q$

q

$\therefore p$

p	q	$p \rightarrow q$	$((p \rightarrow q) \wedge q) \rightarrow p$

Examples

- Check the validity of these arguments using the truth table.

$$\begin{array}{l} p \\ \hline \therefore\ p \vee q \end{array}$$

p	q	$p \vee q$	$p \rightarrow (p \vee q)$

$$\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ r \\ \hline \therefore\ p \vee q \end{array}$$

p	q	r	$p \rightarrow q$	$q \rightarrow r$	$p \vee q$	$\big((p \rightarrow q) \wedge (q \rightarrow r) \wedge r\big) \rightarrow (p \vee q)$

Technique #3 (Rules of Inference)

- Technique #1 & #2 can be tedious especially for argument forms involves many propositional variables, i.e. 10 propositional variables. We need to use a truth table to show this argument form is valid which requires $2^{10} = 1024$ different rows.
- Fortunately, we do not have to use truth tables. Instead, we can first establish the validity of some relatively simple argument forms, called **rules of inference**.
- These rules of inference can be used as building blocks to construct more complicated valid argument forms.

Table : Rules of Inference

Rule of Inference	Tautology	Name
p $p \to q$ $\therefore\ q$	$(p \wedge (p \to q)) \to q$	Modus ponens
$\sim q$ $p \to q$ $\therefore\ \sim p$	$\big(\sim q \wedge (p \to q)\big) \to \sim p$	Modus tollens
$p \to q$ $q \to r$ $\therefore\ p \to r$	$((p \to q) \wedge (q \to r)) \to (p \to r)$	Hypothetical syllogism
$p \vee q$ $\sim p$ $\therefore\ q$	$\big((p \vee q) \wedge \sim p\big) \to q$	Disjunctive syllogism

Table : Rules of Inference

Rule of Inference	Tautology	Name
p $\therefore\ p \vee q$	$p \rightarrow (p \vee q)$	Addition
$p \wedge q$ $\therefore\ p$	$(p \wedge q) \rightarrow p$	Simplification
p q $\therefore\ p \wedge q$	$(p \wedge q) \rightarrow (p \wedge q)$	Conjunction
$p \vee q$ $\sim p \vee r$ $\therefore\ q \vee r$	$\big((p \vee q) \wedge (\sim p \vee r)\big) \rightarrow (q \vee r)$	Resolution

Using Rules of Inference to Build Arguments

- Example 1. Show that an argument with premises $\sim p \wedge q$, $r \rightarrow p$, $\sim r \rightarrow s$, and $s \rightarrow t$ and conclusion t is valid.
  - We construct an argument to show that our premises lead to the desired conclusion as follows.

Step		Reason
(1)	$\sim p \wedge q$	Premise
(2)	$\sim p$	_____ using (1)
(3)	$r \rightarrow p$	Premise
(4)	$\sim r$	_____ using (2) and (3)
(5)	$\sim r \rightarrow s$	Premise
(6)	s	_____ using (4) and (5)
(7)	$s \rightarrow t$	Premise
(8)	t	_____ using (6) and (7)

Note that we could have used a truth table to show the validity. However, because we are working with five propositional variables, p, q, r, s, and t , such a truth table would have 32 rows.

Using Rules of Inference to Build Arguments

- Example 2. Show that an argument with premises $p \to q$, $\sim p \to r$, and $r \to s$ and conclusion $\sim \mathrm{q} \to \mathrm{s}$ is valid.
  - We construct an argument to show that our premises lead to the desired conclusion as follows.

Step	**Reason**
(1) $p \to q$	Premise
(2) $\sim q \to \sim p$	_____ of (1)
(3) $\sim p \to r$	Premise
(4) $\sim q \to r$	_____ using (2) and (3)
(5) $r \to s$	Premise
(6) $\sim \mathrm{q} \to s$	_____ using (4) and (5)

Fallacies (เหตุผลวิบัติ)

- Several common fallacies arise in incorrect arguments.
- These fallacies resemble rules of inference but are based on contingencies rather than tautologies.
- The proposition $\big((p \to q) \land q\big) \to p$ is not a tautology, because it is false when p is false and q is true. However, there are many incorrect arguments that treat this as a tautology. This type of incorrect reasoning is called **the fallacy of affirming the conclusion**.
- The proposition $\big((p \to q) \land \sim p\big) \to \sim q$ is not a tautology, because it is false when p is false and q is true. Many incorrect arguments use this incorrectly as a rule of inference. This type of incorrect reasoning is called **the fallacy of denying the hypothesis**.

Examples of Fallacies

- Example of the fallacy of affirming the conclusion.
  - This argument is invalid.
    - If you do every problem in this book, then you will learn discrete mathematics. You learned discrete mathematics. Therefore, you did every problem in this book.
    - $\big((p \to q) \land q\big) \to p$

- Example of the fallacy of denying the hypothesis.
  - This argument is invalid.
    - If you do every problem in this book, then you will learn discrete mathematics. You don't do every problem in this book. Therefore, you will not learn discrete mathematics.
    - $\big((p \to q) \land \sim p\big) \to \sim q$

Rules of Inference for Quantified Statements

Rule of Inference	Name
$\forall x\, P(x)$ $\therefore\ P(c)$	Universal instantiation
$P(c)$ for an arbitrary c $\therefore\ \forall x\, P(x)$	Universal generalization
$\exists x\, P(x)$ $\therefore\ P(c)$ for some element c	Existential instantiation
$P(c)$ for some element c $\therefore\ \exists x\, P(x)$	Existential generalization

Rules of Inference for Quantified Statements

- **Universal instantiation** is the rule of inference used to conclude that $P(c)$ is true, where c is a particular member of the domain, given the premise $\forall x\ P(x)$.

- **Universal generalization** is the rule of inference that states that $\forall x\ P(x)$ is true, given the premise that $P(c)$ is true for all elements c in the domain.

- **Existential instantiation** is the rule that allows us to conclude that there is an element c in the domain for which $P(c)$ is true if we know that $\exists x\ P(x)$ is true.

- **Existential generalization** is the rule of inference that is used to conclude that $\exists x\ P(x)$ is true when a particular element c with $P(c)$ true is known.

Example 1

Show that the premises "Everyone in this discrete mathematics class has taken a course in computer science" and "Marla is a student in this class" imply the conclusion "Marla has taken a course in computer science."

Solution: Let $D(x)$ denote "x is in this discrete mathematics class,"

$C(x)$ denote "x has taken a course in computer science."

Then the premises are $\forall x\ (D(x) \rightarrow C(x))$ and $D(Marla)$. The conclusion is $C(Marla)$.

The following steps can be used to establish the conclusion from the premises.

Step	**Reason**
(1) $\forall x\ (D(x) \rightarrow C(x))$	Premise
(2) $D(Marla) \rightarrow C(Marla)$	_____ from (1)
(3) $D(Marla)$	Premise
(4) $C(Marla)$	_____ from (2) and (3)

Example 2

Show that the premises "A student in this class has not read the book," and "Everyone in this class passed the first exam" imply the conclusion "Someone who passed the first exam has not read the book."

Solution : The premises $\exists x\ (C(x) \wedge \sim B(x))$ and $\forall x\ (C(x) \rightarrow P(x))$ lead to the conclusion $\exists x\ (P(x) \wedge \sim B(x))$.

These steps can be used to establish the conclusion from the premises.

Step	**Reason**
(1) $\exists x\ (C(x) \wedge \sim B(x))$	Premise
(2) $C(a) \wedge \sim B(a)$	_____ from (1)
(3) $C(a)$	_____ from (2)
(4) $\forall x\ (C(x) \rightarrow P(x))$	Premise
(5) $C(a) \rightarrow P(a)$	_____ from (4)
(6) $P(a)$	_____ from (3) and (5)
(7) $\sim B(a)$	_____ from (2)
(8) $P(a) \wedge \sim B(a)$	_____ from (6) and (7)
(9) $\exists x\ (P(x) \wedge \sim B(x))$	_____ from (8)

Combining Rules of Inference for Propositions and Quantified Statements

Rule of Inference	Name
$\forall x\ (P(x) \rightarrow Q(x))$ $P(a)$, where a is a particular element in the domain $\therefore\ Q(a)$	universal modus ponens
$\forall x\ (P(x) \rightarrow Q(x))$ $\sim Q(a)$, where a is a particular element in the domain $\therefore\ \sim P(a)$	universal modus tollens

Example

Assume that "For all positive integers n, if n is greater than 4, then n^2 is less than 2^n" is true. Use universal modus ponens to show that $100^2 < 2^{100}$.

Solution: Let $P(n)$ denote "$n > 4$" and $Q(n)$ denote "$n^2 < 2^n$".

We are assuming that $\forall x\ (P(n) \rightarrow Q(n))$ is true.

Note that $P(100)$ is true because $100 > 4$.

It follows by universal modus ponens that $Q(100)$ is true,

namely that $100^2 < 2^{100}$.